

Abstract

Malaria, dengue, chikungunya, and Zika are among the mosquito-borne illnesses that continue to be a major global public health concern, especially in tropical and subtropical areas. Every year, millions of people are impacted by diseases spread by mosquitoes. We became aware of the severe health and financial impact these illnesses have on various communities while working on this project. In addition to posing health and environmental risks, we found that traditional mosquito control techniques like chemical repellents and insecticides gradually lose their effectiveness as mosquitoes become resistant. This inspired us to look into a more dependable and safe solution.

In this project, we created an automated Smart Mosquito Elimination System that offers a totally chemical-free and energy-efficient method. We created a dual-attraction mechanism during the first design stage that combines yeast-based CO₂ simulation with 365 nm UV-A light. When compared to a single attractant, it was found that this combination attracted mosquitoes more successfully.

Every functional module in the system is coordinated by an ESP32 microcontroller. During testing, we employed a UV-A LED array for visual appeal and an ultrasonic humidifier to mimic human respiration through controlled mist generation. As we have already seen, the DC suction fan was successful in channeling the mosquitoes towards the trapping chamber where the closed high-voltage electric mesh, running at around 2500V, eliminated the mosquitoes. Comparing the conventional approach with the proposed multi-modal method, the latter was seen to have a significant impact on the attraction and annihilation of mosquitoes during experimentation. As we have already seen, the proposed system's power consumption was below 10W, thus making it possible for it to operate continuously.

On the whole, looking back at what was created, the prototype developed was a secure, cost-effective, and environmentally friendly way of controlling mosquitoes. In this project, it was discovered that by using biological attraction principles and embedded systems, it is possible to come up with a useful result beyond what is possible by the current means of controlling mosquitoes. The system can be used in public and residential areas where the reduction of the spread of diseases caused by mosquitoes is a major concern.

Keywords: Smart Mosquito Elimination System, ESP32, UV-A LED, CO₂ Simulation, Ultrasonic Humidifier, Embedded Systems

CHAPTER 1

Introduction

1.1 Introduction

However, the threat of mosquito-borne diseases still looms over the world. As we were working on this project, we came to understand the impact of the severity of these diseases on the world at large. Every year, millions of people across the globe are infected by diseases such as dengue, chikungunya, Zika, etc., and malaria. Based on the research we did on the topic, we came to the conclusion that the impact is more severe in tropical and subtropical regions.

In examining this issue, we also discovered that contemporary issues are perpetuating the situation. Growth in cities, especially with stagnant water levels in ponds, ditches, and crops due to urbanization, as well as changing weather patterns, are causing a greater number of mosquitoes that spread diseases faster. They are not just environmental issues, as they are also impacting the number of outbreaks, especially in highly populated areas

During our research on the already existing solutions, we learned that most homes still use chemical repellents, insect coils with insecticides, and electronic vaporizers. Our research showed that these solutions provide temporary relief for a short period. When these products are used consistently over time, mosquito resistance to these products develops, making them useless for that purpose. In addition, these products release toxins into the environment that are bad for human health, especially for kids and older people. These products are also not the best option for a long-term solution because they are also a continuous expense.

These constraints prompted us to do things differently for this project. The solution to this problem is the automated smart mosquito elimination system, which uses biological optical attraction techniques without the aid of any poisonous chemicals. The coloration of this system is achieved by the use of yeast-based CO₂ emulation with a 365 nm UV-A light source, which was established during tests to be a good attractor or lure for mosquitoes and safe for the human race and the environment. ESP32 microcontroller controls all the modules with the help of a buzzer and has taken action on all modules. We have also been able to operate the system in real time with low-power hardware and have been able to operate the system below 10W so that it could be deployed outside a non-laboratory setup.

1.2 Background and Motivation

What we learned about mosquitoes] One of the most fascinating things we discovered early on when starting to work on this project is that the behavior of mosquitoes is related to a number of various cues from the environment. First and foremost, we discovered that mosquitoes use various cues such as carbon dioxide, thermal signatures, humidity, and certain wavelengths of light to locate their hosts. Among the cues we discovered is the use of carbon dioxide. As a result of the fact that carbon dioxide is a byproduct of human respiration, carbon dioxide plays an extremely crucial role as a cue en route to the host. Realization which was an important part of our system design. In the course of conducting a literature study, we also came to understand that certain wavelengths of light, particularly those in the UV-A band around 365 nm, can attract species of mosquitoes such as Anopheles and Culex. This gave us the idea that the use of both the simulation of carbon dioxide and the use of UV-A light would improve the efficiency of attraction compared to the use of a single method. Upon reviewing existing mosquito control solutions,

we found that many systems use CO₂ cylinders, chemical attractants, or large trapping devices. We found that these systems have several disadvantages. They are expensive and require frequent maintenance, thus are inappropriate for routine residential applications. Furthermore, many of these systems do not have an intelligent control scheme and more energy is consumed than it should be. We also found that many of these systems rely on a single-mode attraction, whereas a mosquito in real environments may encounter multiple stimuli simultaneously. A single-mode attraction scheme may not attract all target mosquitoes. This is one of the reasons why many existing devices do not perform well in real situations.

We were inspired to create a miniature, low cost and chemical free mosquito elimination system. Instead of using CO₂ cylinders, we used an ultrasonic mist generation method to mimic the human respiration in a safe and inexpensive way. During our tests, we found that this method was effective enough to produce good attraction results without expensive equipment.

This system employs a multi-modal system that consists of a UV-A light and airflow guidance using a DC fan and a high voltage electric mesh for elimination. All of these modules are controlled by an ESP32. We found that using these modules in combination during our tests made the performance of the system better. Each module also helps the others.

In this work, we also aimed at simplicity, safety, and ease of use: we designed a device that can be used at home or in an outdoor-like setting without technical knowledge or continuous maintenance. Generally speaking, the proposed approach provides a practical and efficient mosquito control with a minimum ecological footprint and complexity. combination enhanced the overall performance because each module benefited the other.

1.3 Problem Statement

Even today, vector-borne diseases remain a significant challenge to public health across the world. As we embarked on this project, we quickly realized the need for safer, more effective and more energy efficient mosquito control methods. In our study, we found that many of the current methods still rely heavily on chemical repellents and insecticides. While these are commonly used, they pose risk to the environment and long term health.

We also observed that many modern electronic mosquito control devices have practical limitations: Some consume a lot of power, while others do not have intelligence or automation, so that they cannot be used as a continuous solution. We also observed that chemical repellents may contribute to indoor air pollution and have a negative impact on respiratory health in closed environments.

To analyze some of the advanced solutions, we studied CO₂-based mosquito traps that use this and other attractants. These systems are adequate. However, they rely on pressurized gas cylinders. According to our observations, this results in bulky, expensive and hard to maintain system. Such a system is highly impractical for a regular household use.

By recognizing these problems, we had a clear understanding of the need for a more practical solution. We decided to develop a smaller embedded system to address these issues. Considering the mentioned observations, we determined four key design requirements.

First, and most importantly, we wanted a wholly chemical-free attraction method. In our experiments we found that UV-A illumination can lure in mosquitoes without any toxicity, while simultaneous simulation of CO₂ increases this effect significantly.

Second, we wanted a cheap way to simulate CO₂. We utilized an ultrasonic humidifier that could simulate human breathing instead of gas cylinders.

While testing this approach, it worked well while maintaining simplicity and affordability in the system.

Third, we were mindful of keeping power usage low. We wanted this to be less than 10W, and it was never even close during continuous operation. User safety was also a concern. The proper insulation and cages were designed, as it uses a high voltage electrical mesh. This ensures it can be used safely within a home environment.

1.4 Objectives of the Project Work

One of the very first things that we did as a team, after we got together and decided we wanted to work on this project, was to define what we wanted to achieve. We wanted to find out what we could achieve, and we wanted it to be achievable, and we wanted it to be related to what we thought was wrong with the current systems for controlling mosquitoes. After a couple of back and forths, we came up with these goals.

The first goal was to design and implement a smart control system that used an ESP32 microcontroller. Based on different possibilities, we selected the ESP32 for our project due to its low power consumption, integrated capabilities, and balanced processing capabilities.

The second goal was to come up with a safe and affordable way of simulating the CO₂. We considered alternatives to the expensive CO₂ cylinders, hence the acquisition of the ultrasonic humidifier to mimic the respiration of humans. The idea was successful in attracting mosquitoes during the simulation, as well as being affordable and simple.

The third goal was to visually draw attention using a UV-A LED array operating at a wavelength of 365 nm. Our research indicated that this wavelength attracts mosquito species. We also discovered that UV-A light enhances attractiveness performance.

Designing an airflow guidance system using a DC suction fan was the fourth goal. We established the need for mosquitoes to be guided to the chamber for capture since attraction alone would not work. The airflow from the fan increased the overall capture efficiency. Another important goal was ensuring user safety. Since the system utilizes a high voltage electrical mesh, we provided proper insulation and enclosures. This was necessary for ensuring proper usage of the system in homes.

Lastly, we paid attention to keeping the power consumption down. Our aim was to keep the system consumption at or below 10W, so that the system could operate continuously without incurring increased costs due to power consumption, which was confirmed by testing the system.

1.5 Organization of the Report

We divided the report into several chapters, each devoted on a particular area of the Smart Mosquito Elimination System, so presenting our findings clearly and logically. Preparing this report revealed that this organized approach assists the reader step by step—from grasping the issue to interpreting the ultimate results and conclusions

Chapter 2 — Literature Review

Chapter 2 provides an in-depth analysis of the current mosquito control and removal systems. We have looked at many of these during our research and realized that, despite the many solutions, there are problems associated with each of these systems. The background required to appreciate the need for the upgraded system is also presented in this chapter

Chapter 3 — Analysis of Existing Technologies

In Chapter 3, we are focused on evaluating different mosquito control methods that are currently available. We evaluated different systems, and we established that they all have common issues, including high power consump-

tion, lack of automation, and use of chemical-based methods. This chapter helped us understand the direction of the intended solution.

Chapter 4 — Proposed System Design and Methodology

This method is discussed in detail in Chapter 4. It entails the system’s architecture, operating principles, specific components, and the process used throughout the development process. It is explained how all modules were developed and integrated into a complete system.

Chapter 5 — Implementation, Testing, and Results

The performance review of the system, along with details on testing, can be found in Chapter 5. During testing, the effectiveness of the multi-modal attraction approach was found to be better than other methods in capturing mosquitoes.

Chapter 6 — Conclusion and Future Work

In Chapter 6, the results of the project are discussed, along with possible changes that can be made in the future. This includes a general overview of the performance of the system and possible improvements. **Program Outcomes, PSOs, and SDGs Mapping**

Lastly, the report offers a section that plots the project results against appropriate Sustainable Development Goals (SDGs), Program Outcomes (POs), and Program Specific Outcomes (PSOs). This illustrates the bigger impact and significance of the project.

1.6 Scope of the Project

One of the things that really thrills us as we were designing this system is that we discovered just how extensive the possibilities are for using our Smart Mosquito Elimination System. From the very start of this project, we’ve designed this system for use in the world, not just as a technological model, but as something that can actually change people’s everyday lives in

many different ways.

1.6.1 Residential Use

The quickest and easiest application we came across was the normal domestic setting. We designed this system to function as a reliable indoor mosquito control system running continuously without the use of a single toxic substance whatsoever. Our system was found to be best suited for bedrooms, living rooms, and even small apartments due to its size and power requirements.

1.6.2 Healthcare Environments

We also saw the benefits that hospitals, clinics, and other healthcare establishments could reap from a chemical-free solution for controlling mosquitoes. The last thing anyone needs is to be exposed to dangerous chemicals from the repellents we use in a place where people may already be vulnerable. The completely chemical-free nature of our system makes it intrinsically safe for both medical staff and patients while providing a clean and chemical-free environment without mosquitoes and the dangers of disease transmission.

1.6.3 Rural and Semi-Urban Communities

The possible impact in the rural and semi-urban areas was one application that felt especially important to us during this project. Rural and semi-urban areas are the places where the maximum mosquito-borne diseases occur, mainly because of stagnant water, poor sewage facilities, and lack of access to expensive control measures. The cost-effectiveness of the design makes it extremely affordable and reachable for the larger segment of the population that needs it the most but is often ignored by expensive business solutions.

1.6.4 Public Spaces

In this project, we also looked at its use in a public place, for example, in a restaurant, outdoors, or even in a park on a larger scale. As we observed from the discussions, real public safety also depends on mosquito control in these environments, aside from convenience. Scalability was the aim in our design, and so a number of units could realistically be used in a larger area for significant coverage.

1.6.5 Technical and Academic Significance

Besides its practical applications, we have also seen the general technical importance of what we had created in this project, as the research showed the way in which the technology of embedded systems, the biological attractor mechanisms, and the power-efficient hardware design came together to give the real world integration of these three components. It was also seen during the development that this kind of synergy, where multiple disciplines come together, is the way forward for engineering solutions to be developed.

We also put the mechanism in place with a view to the future, keeping growth in mind. It was also discovered in the process of this project that the ESP32 microcontroller, which we chose for the project, provided a good base for further improvements, which could be made in the future, for example, in the form of IoT-based remote monitoring, cell phone control, and even a completely automatic smart home integration, which could make the system stronger in the future.

1.6.6 Overall Scope

We also reminded ourselves throughout this process that, in our design, we were not simply building a basic mosquito trap. Instead, we were aiming for something far greater: a solution that was not only ecologically friendly,

scalable, and practical but also appropriate for adaptation and deployment for a wide range of scenarios. As we traveled throughout our journey, we realized that all our design decisions, no matter how trivial, were eventually instrumental in achieving our ultimate goal.

CHAPTER 2

Literature Survey

2.1 Introduction

One of the first things we did in starting this project was studying the current research and methods of mosquito control. In this field, we found that the global incidence of mosquito-borne diseases, which include malaria, dengue, chikungunya, and Zika, has led to increased research [1]. In particular, in tropical and subtropical regions where the environment is conducive for mosquito breeding, these diseases are still a major health issue.

2.1.1 Traditional Chemical-Based Approaches and Their Limitations

First, we considered some of the frequently used techniques for controlling mosquitos, which include chemical repellents and pesticides. Though this method is frequently used, some of the disadvantages that this method has in the long term include environmental pollution, increased resistance among mosquitos, and health risks to humans. This method also has the drawback that, since mosquitos constantly need to be repelled, they become expensive in the end and become less effective.

2.1.2 Emergence of Alternative Mosquito Control Technologies

Due to these constraints, other methods have also been explored by scientists. These include UV light traps, CO₂-based attractants, pheromone-based

methods, and IoT-based monitoring devices, among others. Unlike the use of chemicals, these methods aim at attracting the mosquitoes using human mimicking signals.

2.1.3 Biological Attraction Techniques

Of these possibilities, the use of organic methods for attracting mosquitoes appears to be a success. Mosquitoes are attracted by certain wavelengths of light, body heat, humidity, and carbon dioxide, among other things. Modern methods could be efficient in attracting mosquitoes by simulating these components in a way that does not involve the use of chemicals. Our system design was largely inspired by this idea.

2.1.4 Challenges in Existing Systems

Despite this, we discovered that some of the systems that are available have certain issues, which include a higher price, increased power consumption, lack of portability, and absence of intelligent control. This reduces the usability of the systems for everyday use.

2.1.5 Purpose of This Chapter

This chapter focuses on the advantages and disadvantages of the available mosquito control methods, as well as a detailed analysis of the same. From this study, a number of research gaps have been identified that are not sufficiently addressed by the available systems. These findings have been used to inform the proposed Smart Mosquito Elimination System that focuses on enhancing safety, convenience, and efficiency.

2.2 Review of Prior Research

During this endeavor, we thoroughly researched the existing body of literature pertaining to the various control methods for mosquitoes. We found that more efficient and safer alternatives for the existing conventional chemical-based methods have already been produced by researchers around the world, It was observed that most of these methods seek to enhance trapping efficiency while wholly avoiding hazardous chemicals by using biological attraction mechanisms that mimic natural human signals—such as carbon dioxide, body temperature, humidity, and particular light wavelengths. Sefat et al.— Multi-Attractant Intelligent Trapping System One of the first research that we came across in this endeavor was the research done by Sefat et al. on the development of a smart mosquito trapping system that utilizes a combination of attractants such as carbon dioxide, ultraviolet rays, heat, and pheromones. We found their results to be particularly intriguing for our project since combining several stimuli produced significantly increased attraction compared to those relying on a single attractant. This discovery suited our theory of designing quite well. It was, however, discovered that their system’s dependence on compressed CO2 cylinders resulted in several practical disadvantages, including increased physical size, increased system cost, and increased maintenance needs. In our evaluation, these aspects make this system much less feasible or even more unrealistic for constant or portable, or even home, use, which is exactly what we aimed to fill in with this project. Mwanga et al. UV LED-Based Mosquito Traps We also evaluated the research done by Mwanga et al. regarding their evaluation of the effectiveness of UV LED-based mosquito traps, especially for species like *Anopheles* and *Culex* [2]. The project’s findings revealed that their study provided robust and convincing evidence of the efficacy of UV light in attracting mosquitoes, both in actual outside conditions and in labo-

ratory settings. This immediately backed our plan to include a 365nm UV-A LED array into our own system. Nevertheless, it was observed that many of the systems assessed in their study ran without any kind of automation or intelligent control even though UV attraction was effective. We noted during our review that this lack of intelligent control greatly restricts UV-based traps' general efficiency and adaptability in real-world deployment situations; another drawback we were dedicated to solve in this study. Mosquito Monitoring and Detection System by Lee et al. Lee et al. proposed another research we investigated during this project; it created a mosquito monitoring system using sensing technologies to find and follow mosquito activity in a specific area [4]. Looking at data collecting and monitoring, we thought this effort was actually amazing. Nevertheless, it was discovered that their system concentrates mostly on monitoring and tracking; it does not directly target mosquito eradication. Our research indicated that while such monitoring systems are undoubtedly useful in research and public health surveys, their practical usage in actual mosquito control situations is limited without a coordinated method of elimination. This project reinforced our original intention to develop a system that combines both strong attraction and elimination. Systems for Controlling Mosquitoes Built on the Internet of Things Finally, as part of this project, we also examined the recent developments in the field of IoT-based mosquito control systems. Most of these systems are seen to be integrating the use of the microcontroller platform, along with the use of the attraction and elimination of mosquitoes via the use of UV light and electric mesh, respectively, to make the systems more intelligent. These new systems appear to be a considerable step forward. It was observed that these IoT-based systems have real enhanced automation capabilities and real-time monitoring capabilities compared to older architectures. Still, as we have assessed these systems, we observed that they have limitations with a constant set of practical issues that arise

with increased electricity usage, increased physical footprint, and maintaining long-term operation reliability. Observed research gaps distilled After carefully evaluating all these research, we have observed that there is a constant pattern with a set of limitations with the current mosquito control methods in this particular project. It was noted that none of the current systems simultaneously offered a mix of these following characteristics: chemical-free operation, power consumption, intelligent automation, portability and smallness of the system, and cheapness for household use. By designing our proposed Smart Mosquito Elimination System to address each of these deficiencies in a realistic and integrated fashion, these aforementioned gaps became the foundation on which our project was built.

2.3 Identified Research Gaps

Evaluating and analyzing existing mosquito control systems identified a number of general restrictions affecting their practical viability. These limitations were identified within a number of methodologies, not just one. The need to specify the course of our proposed solution was aided by the awareness of such limitations.

2.3.1 Dependence on CO₂ Cylinders

One of the main challenges we encountered with these advanced mosquito traps is their dependency on pressurized CO₂ tanks. Although CO₂ is an attractive lure for mosquitoes, there are some challenges with gas cylinders. These challenges include increased costs, constant need to refill gas, and decreased portability due to increased volume. This makes these traps unpractical for use by households. To solve this challenge, we used a cost-effective and safer alternative for simulating CO₂: ultrasonic mist generating method.

2.3.2 Use of Chemical-Based Attractants

Yet another important limitation is the use of chemical attractants and repellents. Even though this may be of some temporary benefit, there are issues of health and environmental integrity that come into play. Especially in the case of indoor usage, this may cause problems of breathing difficulties. Moreover, contributing to environmental degradation is the widespread use of these chemicals. These results prompted us to devise a completely chemical-free system.

2.3.3 Lack of Intelligent Control Mechanisms

We also found out that the existing electronic mosquito control systems lack sophisticated control. This means that the fixed systems do not change according to the environment. This has resulted in wastage of power and inefficient running of the systems. To address this challenge, we incorporated an ESP32 microcontroller to improve the coordination of the system components and increase efficiency.

2.3.4 Overall Analysis

From our study, it became clear that these limitations are linked and should be handled together. Every one of these voids impacted our creative choices. Our goal in designing a more workable and effective mosquito eradication system was on chemical-free operation, affordable deployment, and smart control.

2.3.5 Comparison of Existing Mosquito Control Systems

After considering different strategies of mosquito control in this project, we concluded that it would be helpful to show them in a neat and structured

approach. Table 2.1 contains a organized comparison of formerly suggested methods for this purpose.

Comparing these approaches let us more clearly grasp their benefits and drawbacks. This comparison also helped us spot gaps shared by current systems and gave a clear reasoning for the design decisions taken in our recommended system.

Table 2.1: Comparison of Existing Mosquito Control Systems

Method	Attraction Type	Advantages	Limitations
CO ₂ Cylinder-Based Traps	Carbon dioxide emission	High attraction efficiency	Expensive, bulky, requires maintenance
UV Light Traps	Ultraviolet light	Low cost, simple design	Limited effectiveness when used alone
Pheromone-Based Traps	Chemical attractants	Species-specific attraction	Environmental concerns, recurring cost
IoT-Based Smart Traps	UV + sensors + control	Automated operation, monitoring	High power consumption, complex design
Proposed System	UV + Mist Simulation + Air-flow	Chemical-free, energy-efficient, compact	Limited coverage area

2.4 Challenges in Existing Systems

Having gone over current mosquito control techniques, we discovered that there were still several realistic difficulties. Though many technologies have been created, none of them offer a solution that is affordable, secure, clever, and suitable for daily usage all at once. Before presenting our solution, it was imperative to grasp these difficulties.

2.4.1 Dependence on CO₂ Cylinders

Relying on pressurised CO₂ cylinders in many commercial mosquito traps presents one of the most difficult problems. Though CO₂ is a great attractant, employing gas cylinders presents logistical problems. These systems are expensive, require continuous refilling, and are not practical for home use. Because of these limitations, they are not widely adopted even though they are effective.

2.4.2 Health and Environmental Risks of Chemical Attractants

Another issue of concern lies in the use of chemically derived attractants and pesticides, which, though might be efficient in the short term, can cause problems in the long term, especially in indoor environments, and also result in environmental pollution. In this respect, we felt the need to come up with a totally chemical-free approach

2.4.3 Lack of Intelligent Control

We also discovered that several of the current systems lack a smart control system. In most of the devices, they operate at a constant rate and do not vary depending on their environment. This leads to wastage of power and poor working habits. The addition of an ESP32 control system helped to address this issue and allow for better working habits.

2.4.4 Single-Mode Attraction Limitation

Existing mosquito traps make use of a single attract strategy, for instance, the use of UV light. However, in a real situation, mosquitoes are influenced by a number of factors. It is possible that systems that make use of a single factor may not work effectively. This understanding prompted our decision to

design a multi-attraction strategy that incorporates the use of UV-A light and simulation of CO₂.

2.4.5 Power Consumption and Device Size

From the problems that were identified in this research, it was concluded that to have a good system, these problems have to be solved at the same time. The constraints that were identified in this research helped to shape the decisions that were made in the design of the small, energy-efficient, and chemical-free mosquito elimination device

2.4.6 Summary of Challenges

From our study, we decided that to create a good system these problems have to be tackled simultaneously. The constraints discovered here shaped our design choices and steered the development of a small, energy-efficient, chemical-free mosquito elimination device.

2.5 Summary of Literature Review

In this chapter, we were able to explore the previous literature and develop a good understanding of the various existing techniques for controlling mosquitoes. By exploring various options, we were able to understand the advantages as well as the disadvantages. This affected the design process.

These various methods include the use of attraction systems based on CO₂, UV-based traps, pheromone-based methods, and the use of IoT-based systems for monitoring. This variety of methods shows the impact of biological attraction on the behavior of mosquitoes. The efficiency of the trapping of mosquitoes is increased by a large factor by the mimicry of the signals of CO₂ and light.

However, the study also had a number of major restrictions. First, a number of successful systems require pressured cylinders of CO₂ and therefore increase the cost and maintenance requirements. In addition, a number of the systems use only a single method of attraction and this limits their application for use in the field where mosquitoes react to a number of different attractants.

Another huge challenge we encountered was the lack of a smart control in some of the existing equipment. Most of these systems are running continuously without adapting to the conditions of the environment, hence making them less efficient and causing high levels of electricity consumption. These systems are not efficient for use on a daily basis due to these limitations.

These observations led us to a conclusion that a small, efficient, automated mosquito eradication technology is certainly needed. The proposed method in this project intends to combine different methods of attraction with the promise of safe and efficient power consumption. This proposed method directly addresses the shortcomings identified in existing literature and presents a more viable solution.

CHAPTER 3

Analysis of Existing Mosquito Elimination Systems

3.1 Introduction

Before developing our proposed system, it was necessary for us to be familiar with the technologies that are currently available. This chapter provides a technical study of the technologies used for the elimination of mosquitoes and traps that are available currently. This helped us in understanding the pros and cons of the available technologies.

3.1.1 Evolution of Mosquito Control Technologies

However, over the years, techniques for mosquito control have changed dramatically from chemical repellents to sophisticated electronic devices. The contemporary techniques for mosquito control include luring them based on biological and environmental factors, followed by mechanical or electrical killing. Understanding this evolution enables us to comprehend which techniques are effective and which ones need improvement

3.1.2 UV Light-Based Attraction Systems

Most of the methods for attracting mosquitoes use ultraviolet (UV) radiation. These devices attract certain types of mosquitoes using a specific kind of light. This kind of light includes the UV-A category. Simple and inexpensive they are! However, the use of UV radiation alone does not show promise for the trapping efficiency required. This is the main reason we had to combine

this with other methods.

3.1.3 CO₂-Based Attraction Systems

The systems are based on CO₂, which attracts mosquitoes like a human. This is because carbon dioxide is a powerful natural attractant. As a result, it is very efficient. The systems are mostly based on compressed CO₂ cylinders. This is expensive and requires frequent maintenance. This makes it not quite ideal for domestic use. We also looked at others.

3.1.4 Pheromone-Based Attraction Methods

Systems using pheromones attract certain kinds of mosquitoes through chemical signals. They perform well under controlled conditions but require the constant replacement of chemical substances and may have negative effects on the environment. This reduces their effectiveness over a long period

3.1.5 Electric Mesh-Based Elimination Systems

High voltage grids, as part of the electric mesh systems, are the ones that zap the mosquitoes. When the mosquitoes come into contact with the mesh, the electric mesh systems are proven to be effective. The effectiveness of the electric mosquito killer, however, depends on the way the mosquitoes are attracted to the device. Bad total efficiency is the result if the method of attraction is poor, even if the mosquito elimination method is effective.

3.1.6 Key Challenges in Existing Systems

Our research also indicated that most of the existing systems use only one technique of attraction, for example, UV light or CO₂. These systems are

not efficient in real conditions, as mosquitoes react to a number of different stimuli

Additionally revealed is our majority of gadgets' absence of clever control. Higher power usage and lower efficiency result from their constant operation without adaptation to environmental factors.

3.1.7 Purpose of This Chapter

This study's insights guided us in clearly defining the needs for our suggested system. Based on these observations, we aimed on creating a low power consumption and safe operating solution combining several attraction mechanisms with clever control.

3.2 Comparative Analysis

System Type	CO ₂ Cylinder	UV Only	Proposed System
Chemical-Free	Yes	Yes	Yes
Low Cost	No	Yes	Yes
Automation	Limited	Limited	Yes
Energy Efficient	Moderate	High	High
Maintenance	High	Low	Low

Table 3.1: Comparison of Existing Systems with Proposed System

3.3 Summary of Analysis

From the thorough study done in this chapter, we noted that while UV-based attraction and CO₂ simulation among other individual mosquito control technologies are effective, no current system realistically and user-friendly combines all needed characteristics. Most systems still have intelligent control

constraints as well as great maintenance and cumbersome design.

These findings led us to concentrate on enhancing current methods rather than totally transforming the system from zero. We aimed to keep good things while meeting the main drawbacks noted in the review.

One significant change in our system is the substitution of CO₂ cylinders for a yeast-based mist simulation technique. This method provides a safe alternative, which reduces the costs, thereby eliminating the need for frequent maintenance. It also ensures the maintenance of good attraction.

We have also integrated the direction of the airflow with the UV-A light, using the DC suction fan. This ensures that the mosquitoes are directed into the trap chamber, hence not attracted into the trap, but directed into the trap chamber, thereby increasing the overall efficiency of the trap.

Smart control utilizes an ESP32 microcontroller. This is yet another important feature. The improved coordination of the device components is a result of this. This, therefore, improves energy efficiency compared to traditional fixed operation devices. We have also given prime importance to safe operation and low power usage. The device is suitable for home use and operates at less than 10W. It is well insulated to operate at high voltage.

The proposed approach combines multiple improvements into a single design. It provides a practical, effective, and eco-friendly solution for mosquito control while addressing the key limitations of existing methods.

CHAPTER 4

Proposed System Architecture and Methodology

4.1 Introduction

The design and methodology of the proposed Smart Mosquito Elimination System are discussed in Chapter 4. The creation of the system by combining the concepts of biological attraction and system design is explained in this chapter. The main aim was to develop a system that is totally chemical-free, energy-efficient, and efficient.

4.1.1 Understanding Mosquito Behavior

We studied the behavior of mosquitoes to find out what attracts them before we began designing our system. It appears that mosquitoes tend to be attracted to carbon dioxide, temperature of human bodies, humidity, and specific light frequencies. Simulating these natural signals in a controlled manner makes it possible to attract mosquitoes without using dangerous compounds. Our system was based on these considerations.

4.1.2 System Architecture Overview

The system is designed around four main functional modules that work together to attract and eliminate mosquitoes effectively:

- Ultrasonic mist generation for simulating humidity and CO₂
- For visual appeal: UV-A LED array 365 nm
- DC Fan for directional airflow towards the capture chamber

- High-voltage electric mesh for mosquito elimination

These modules are interconnected and operate as a single system, ensuring effective mosquito attraction and elimination.

4.1.3 ESP32-Based Control System

The central system control is provided by the ESP32 microcontroller. It ensures synchronized operation, controlling all module operation. The use of ESP32 facilitates automated control, efficient power management, and coordination of several system components.

4.1.4 Multi-Modal Attraction Strategy

The proposed system utilizes a multi-modal attraction method by employing CO₂ simulation and UV-A light. The multi-modal method is more efficient since mosquitoes react to multiple factors in a natural environment. The presence of two or more attractants enhances the possibility of capturing mosquitoes regardless of environmental conditions.

4.1.5 Safety and Energy Efficiency

Other important design considerations were power efficiency and safety. In order for the high-voltage electric mesh not to accidentally come in contact with anything, it is kept within an insulated framework. With a power consumption of less than 10W, the system can be used continuously in homes.

4.1.6 Purpose of This Chapter

The system design and approach are explained thoroughly in this chapter. It explains the contribution of each component to the system for the effective and safe elimination of mosquitoes through its overall function

4.2 Overall System Architecture

With extensive planning and numerous design iterations, we ultimately determined an overall system design consisting of the following main components, which were ultimately chosen, tested, and implemented as a whole:

- Central Control Unit of ESP32 Microcontroller
- Operating at a 365 nm wavelength, UV-A LED array
- Ultrasonic Humidifier for CO₂ Modeling
- Mechanism of DC Suction Fan
- High- Voltage Electric Mesh: approximately 2500 V
- Voltage Regulation Circuit and AC–DC Power Supply

As the main processing core controlling all functional modules running in concurrent streams, the ESP32 Microcontroller sits at the heart of everything that we created in this project. In fact, the ESP32 Microcontroller handled activation timing, power optimization, and module synchronization surprisingly well in this project. As a matter of routine, it was observed that the ESP32 ensured that everything was running in a safe, efficient, and completely coordinated manner, which was something that we soon came to realize was vitally important for the performance of everything in the system.

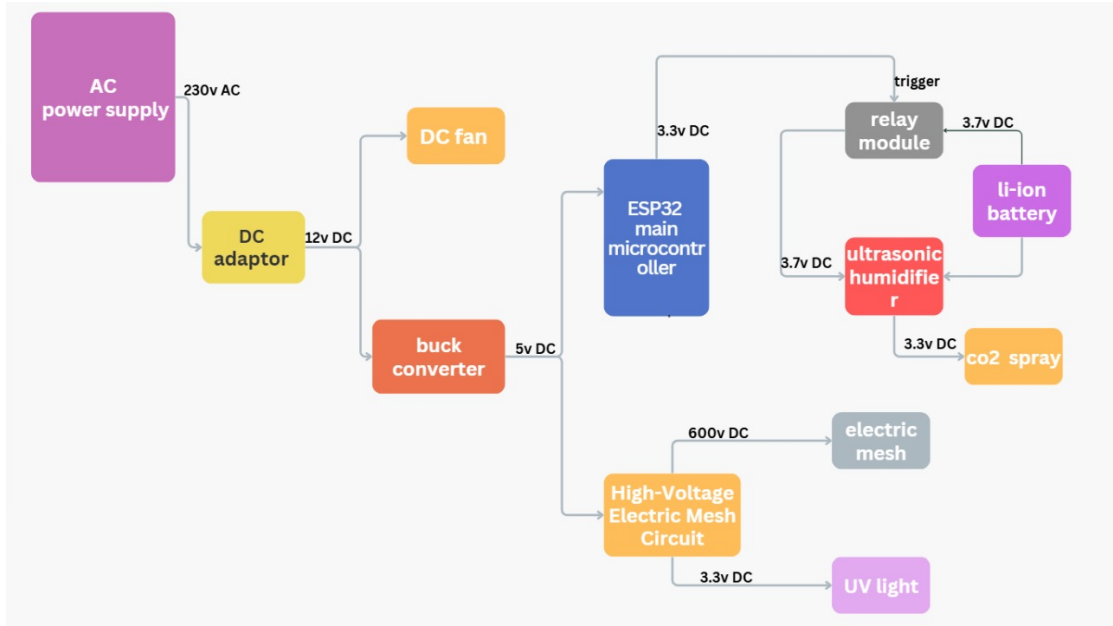


Figure 4.1: Block Diagram of the Proposed Smart Mosquito Elimination System

4.3 Working Principle

First, studying the behavior of mosquitoes will help us before completing the hardware design. We found that mosquitoes are attracted to certain biological markers, which are body heat, carbon dioxide, humidity, and certain light frequencies. Our system design was based on this knowledge

The proposed system works through a multi-stage process, where each stage helps to attract, guide, and eliminate mosquitoes. The working mechanism is explained below:

4.3.1 Stage 1: CO₂ and Humidity Simulation

When the system is turned on, ultrasonic humidifiers emit very little water. This fog mimics the effects of human breathing by generating CO₂-like and humidity signals. Testing revealed this phase draws mosquitoes from a distance without resorting to chemical agents or gas cylinders.

4.3.2 Stage 2: UV-A Optical Attraction

Concurrent UV-A LED array releases light at 365 nm. This wavelength is known to be very attractive to mosquito species. Comparing only one method, the combination of fog and UV-A light raises attraction efficiency.

4.3.3 Stage 3: Airflow Guidance

A DC fan creates regulated airflow toward the trapping chamber once mosquitoes start to approach the system. This airflow steers mosquitoes into the device and stops their exit. We found this phase to be vital for increasing capture efficiency.

4.3.4 Stage 4: Electric Mesh Capture

Mosquitoes are directed toward the high-voltage electric mesh inside the chamber by the airflow. The main system elimination element is this mesh.

4.3.5 Stage 5: Instant Elimination

When mosquitoes come into contact with the electric mesh, they are immediately killed. To ensure safety, the mesh is housed in an insulated framework and operates at about 2500V.

4.3.6 Effectiveness of Multi-Modal Approach

The system's overall efficiency is the result of several attraction strategies working together. Better performance than single-mode systems is achieved because each phase supports the others. This multimodal approach ensures better mosquito capture in real-world situations.

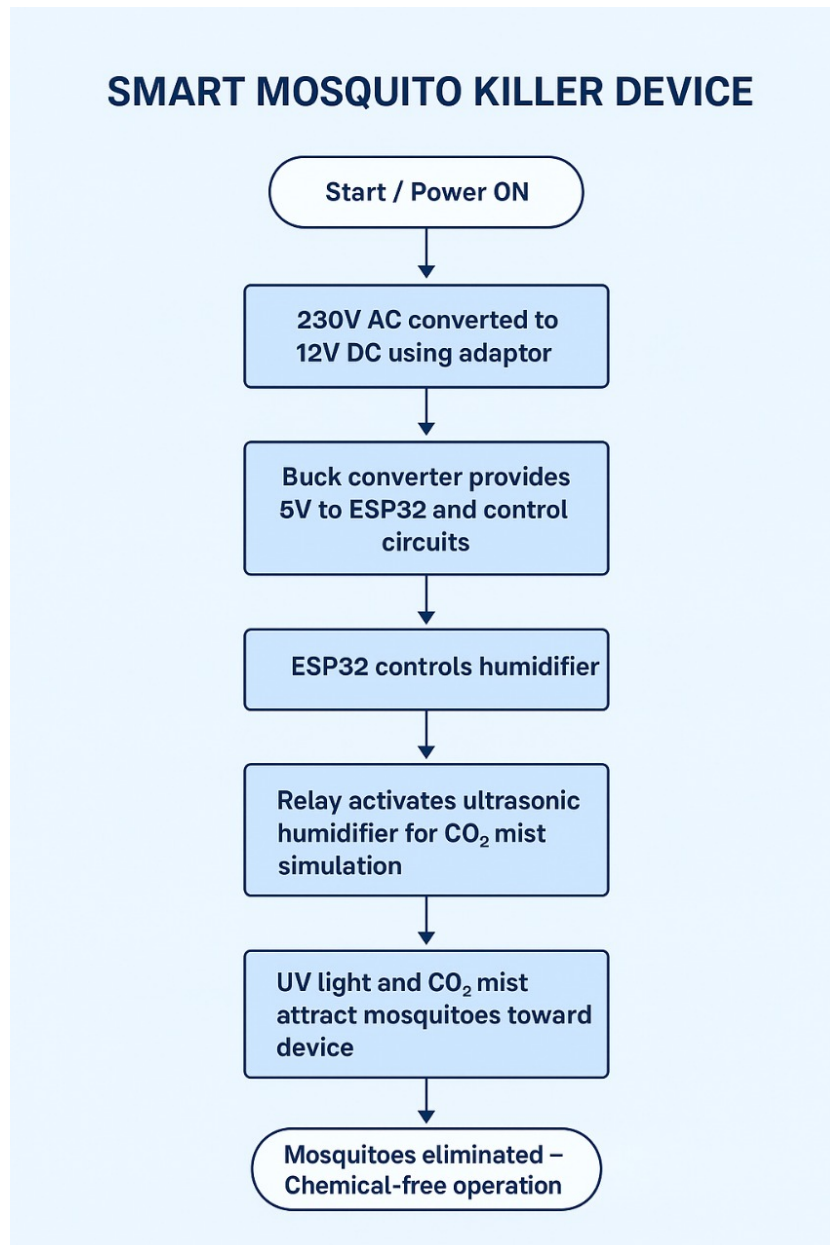


Figure 4.2: Flow Diagram of System Operation

4.4 Block Diagram Description

During the design phase, we made a block diagram to illustrate the interactions between various subsystems of the Smart Mosquito Elimination System. The four main stages of the system are depicted in Figure 4.1: power, control, attraction, and elimination. This well-organized depiction ensured proper module integration and made it possible for us to understand

the activity's flow.

4.4.1 Input Stage – Power Supply

Starting with the power supply unit, which converts 230V AC mains into regulated DC voltages required for different components, the system's proper operation depends on a consistent power source because fluctuations may impact sensitive components like the ESP32 microcontroller and UV-A LEDs. The overall system performance was enhanced by ensuring reliable power distribution.

4.4.2 Control Stage – ESP32 Microcontroller

The main microcontroller of the system is ESP32, which controls all other modules, including the DC fan, ultrasonic humidifier, and UV-A LED array. Improved synchronization and power management are achieved through centralized control, making it more reliable.

4.4.3 Attraction Stage – UV-A LED and Mist Generator

Two processes: UV A illumination and ultrasonic mist creation combine during the attracting phase. Esthetically appealing to the mosquitoes, the UV A LED emits 365 nm light. At the same time, the mist generator simulates humidity and CO2 levels, thus replicating human breathing. Compared to single-mode systems, the combination of these two approaches dramatically increases attraction effectiveness.

4.4.4 Elimination Stage – Electric Mesh

Mosquitoes come into touch with the high-voltage electric mesh once they are lured and directed into the chamber. Generating roughly 2500V guarantees

instant elimination using a boosting circuit. To preserve user safety, the mesh is covered in an insulated structure.

4.4.5 Modular System Design

The modular approach used in designing the system means that every phase runs alone. This simplifies the creation, testing, and maintenance of the system. It also increases flexibility and scalability by letting future upgrades or changes be made without compromising the whole system.

4.5 Power Optimization Strategy

We vowed from the beginning of this effort to always keep total system power consumption under 10W. In this endeavour, we discovered that this was more than just a technical objective; it was a fundamental design philosophy that influenced each and every component selection and execution decision we made throughout the development process. It was discovered that a system that functions well technically but is just too costly to run continuously in a real residential setting could easily result from an unplanned and methodical approach to power optimisation. During this project, we used power optimisation in five specific domains, all of which greatly aided in achieving our overall energy efficiency goal:

- **Selecting UV-A LEDs with Low Power** The careful selection of low-power UV-A LEDs for our optical attraction array was one of the first power-related decisions we made for this project. We found that the power requirements of UV LEDs vary greatly during our component research; selecting the appropriate specification allowed us to achieve a powerful and efficient 365 nm light output while consuming the least amount of energy possible. Throughout testing, we observed that these

LEDs maintained consistent attraction performance without significantly increasing the overall energy budget.

- **Energy-Saving Direct Current Fan** An energy-efficient DC suction fan was another component of our airflow control system. Unselected fan motors can be among the most energy-intensive parts of any embedded system, it was found early on in this project. We discovered that selecting a DC fan with the ideal ratio of power efficiency to airflow performance allowed us to maintain effective mosquito guidance into the elimination chamber without materially compromising our 10W target. Testing revealed that within our power budget, the fan consistently and reliably performed very well.
- **Controlled Duty Cycle Ultrasonic Humidifier** Controlled duty cycling of the ultrasonic humidifier was maybe the most imaginative power optimisation we applied on this project. Early tests revealed that running the humidifier at full, continuous level was pointless; mosquito attraction cues do not need to be constant to be efficient. Through the ESP32, we incorporated a duty cycle approach alternating the humidifier between active and rest times in a regulated rhythm in this project. Testing revealed that although this technique greatly lowered the share of the humidifier in total energy consumption, it also kept excellent CO₂ and humidity simulations.
- **Effective AC-DC Voltage Control** We also focused on in this project how effectively we transformed the 230V AC mains supply into the regulated DC voltages our components needed. During the design phase, it became clear that before any functional component consumes even one watt, an ineffective power supply module can waste a surprisingly high amount of energy as heat. We used an effective AC-DC regulating system that

lowered conversion losses and provided all modules with steady, clean voltages. We saw during testing that this significantly helped us to maintain our overall system power consumption far inside our target.

- **Operating Low-Power ESP32 Microcontroller** At last, we totally made use of the built-in low-power operating features of the ESP32 microprocessor. During implementation, it was discovered that the ESP32 offers several power management features that, when used properly, can significantly reduce the controller's energy footprint during times of consistent or frequent operation. The ESP32 efficiently managed all system coordination tasks during continuous operation testing, consuming only a small portion of our total power budget.

Overall Power Efficiency In this project, the combined effect of all five of these optimisation strategies was exactly what we had hoped for. During extended periods of continuous operation, testing revealed that the system consistently operated significantly below our 10W target. Because mosquitoes are most active during the night and early morning, when people are asleep and most vulnerable, this level of energy efficiency was found to make our Smart Mosquito Elimination System truly feasible for 24/7 uninterrupted functioning. Finding a way to achieve this goal without significantly compromising performance was one of our most satisfying technical achievements in this project.

4.6 Safety Mechanisms

We gave safety top priority from the start of this project. We learned that handling high-voltage components at home should not be done carelessly because even small safety design errors can have significant consequences. As a result, we made sure that the following preventive measures were applied

carefully and precisely:

- **Complete High-Voltage Mesh Cover** The electric mesh was housed in a completely insulated chamber that we constructed and operated at about 2500V. It was found in tests that this totally removed any chance of unintentional contact with kids or users.
- **Appropriate metal construction grounding** Here, we properly grounded all of the system's metal parts. Without it, we discovered that any unforeseen fault current could pass through the outer structure and pose a shock risk.
- **Electrical Segregation of Control and High-Voltage Areas** It was evident that the high-voltage area and the ESP32 control circuit were electrically isolated. It was discovered that this division ensures that a defect in one component won't jeopardise or compromise the other..
- **Outer Casing with Protective Properties** This project had us creating a protective outer layer around the whole gadget. Normal handling and everyday use, we found, provide an extra layer of physical protection.
- **Stable Voltage Boosting Circuit** To avoid spikes or other unwanted voltage fluctuations, we also added a stable voltage boosting circuit. During testing, we found that a dependable consistent voltage output was sustained during ongoing operation.

We also discovered that all of these things taken together make it a very safe system for unattended use in homes, which we thought was just as important as making it work

4.7 Advantages of Proposed System

- An entirely chemical-free method for the eradication of mosquitoes was employed; the results showed this had no negative impact on the environment and the population living within the area.
- Compact and portable design
- Low maintenance requirements
- Through the embedded intelligence we designed, the system was able to operate in a totally automated and regulated manner without the need for human involvement whatsoever
- In this project, we have seen that every choice, starting from the selection of the components to the power management, has helped us to create something really green and useful for the real world.

4.8 Hardware Components Description

4.8.1 ESP32 Microcontroller

The brain of our whole system in this project was ESP32. We employed it as the main processing unit responsible for overseeing all activities of all hardware components, including the UV-A LED array, ultrasonic humidifier, DC fan, and high voltage mesh. We tested it and established that it executes all these tasks concurrently without any issues, thus ensuring seamless and synchronized execution of all tasks.



Figure 4.3: ESP32 Microcontroller Module

After considering a number of possibilities for a microcontroller, ESP32 was chosen, especially for this project. It was realized that it offered precisely what was needed: low power consumption, sufficient GPIO pins for connecting all our devices, and reliable performance even when used non-stop. It was also discovered that the ESP32 has Bluetooth and Wi-Fi capabilities, which only increased our enthusiasm for this particular project. While we didn't include this as part of our current prototype, it was noted that this will allow the system to be really future-ready for IoT-based remote monitoring and smart home integration.

4.8.2 UV-A LED Module

In this project, we have employed a UV-A LED array, which was used to emit light at a precise wavelength of 365nm, a wavelength we have tested and proved quite successful for attracting mosquitoes visually. Testing showed that mosquitoes would consistently and rapidly be attracted to this wavelength.



Figure 4.4: UV-A LED Array for Mosquito Attraction

Especially, these particular LEDs were chosen for their energy efficiency, as well as for their aesthetically pleasing characteristics. As we have noticed throughout this particular project, they maintained a high level of illumination while consuming very little power, which was very important for our case since from the very beginning, we had an unwavering goal of keeping the entire system below 10W.

4.8.3 Ultrasonic Humidifier

However, this was one of the most enjoyable parts of the whole project for us. We decided to use an ultrasonic humidifier to replicate human breathing by creating fine mist instead of using pricey and cumbersome CO₂ cylinders that most current systems use. It was observed that mosquitoes reacted to the mist created by the humidifier in much the same way that they normally would to CO₂ and moisture emanating from a real human host - which was quite interesting to see



Figure 4.5: Ultrasonic Mist Generation for CO₂ Simulation

This method proved to be highly effective. It completely eliminated the demand for compressed gas cylinders, which made our system noticeably less expensive, safer, and much more useful for regular household usage.

4.8.4 DC Suction Fan

We realized early on in this project that something more than mere attraction was required; something to lure the mosquitoes into the trap, as it were. It was for this reason that the DC suction fan was installed, as it was realized that the airflow generated by this fan would provide a reliable route that would direct the mosquitoes toward the elimination chamber, as tests showed that the efficiency of capturing mosquitoes was greatly improved

by the inclusion of this fan, as compared to previous experiments that were carried out without this fan



Figure 4.6: DC Suction Fan

4.8.5 High Voltage Electric Mesh

This is where the mosquito finds its death. As our main tool for elimination, we used in this project a high-voltage electric mesh, which runs on approximately 2500V. As a result of testing, we found out that as long as a mosquito makes contact with the mesh, it gets eliminated instantly. We were also quite conscientious about safety, as we completely insulated a chamber around the mesh so that there is no possibility of unintentional touch throughout regular use.

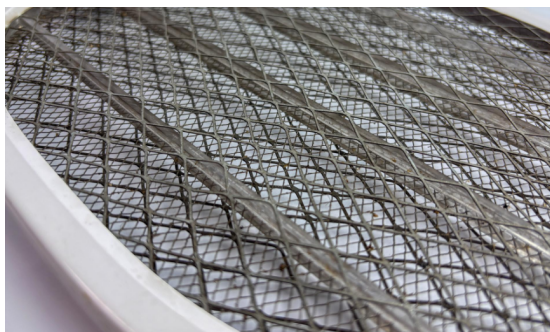


Figure 4.7: High Voltage Electric Mesh

4.8.6 Power Supply Unit

For this particular project, we used a separate power supply unit that converted the input voltage from the 230V AC supply into a stable regulated DC voltage that our components needed. It was evident that the components' performance was being affected by any voltage variation, and a reliable voltage supply was thus very important. It was also evident that our voltage regulation was keeping things running smoothly, all within a power consumption well within our target of 10W.



Figure 4.8: Power Supply Circuit

4.9 Software Algorithm and Control Logic

4.9.1 System Operation Algorithm

The ESP32 control algorithm ensures all parts of the system operate together in a smooth and efficient manner. The ESP32 control algorithm ensures all parts of the system are running smoothly by turning them on in the correct order.

Testing and improvement showed that the following sequence is best for attracting and killing the mosquitoes:

1. System Initialisation

The ESP32 initializes all the devices that are connected when the system is turned ON. Correct initial setup is necessary for steady functioning, without any unexpected behavior.

2. UV-A LED Activation

First, the UV-A LED array is turned on for aesthetic purposes in the scene. This allows the mosquitoes to react to the system from the early stages

3. Humidifier Activation

The ultrasonic humidifier is then turned on for CO₂ simulation mist production. This enhances the effectiveness of the mosquito attraction process.

4. DC Fan Activation

The DC suction fan turns on to produce air flow in direction of the trapping chamber. This increases capture rate and guides mosquitoes into the system.

5. Continuous Monitoring and Synchronization

The ESP32 checks every module constantly to make sure they run in time with one another. This stops performance problems brought on by poor coordination.

6. Power Management

Using optimization methods, the system keeps low power usage by making sure that during constant use the operation stays below 10W.

7. Continuous Loop Operation

The whole procedure operates in a loop that lets automatic and continuous mosquito attraction and elimination occur without human involvement.

Several rounds of testing helped to improve this method. The general performance of the system was shown to depend much on the sequence and pace of every step.

4.9.2 Flow Description

The system runs in a loop continuously whereby several modules run at once. Compared to sequential activation, this concurrent functioning boosts general efficiency.

While the ultrasonic humidifier produces small mist to mimic human breathing, the UV-A LED array offers constant visual appeal. Concurrent with this, the DC suction fan produces constant airflow towards the collecting chamber. These three elements work together to create a great multi-modal environment of attraction.

The airflow guides a mosquito towards the trapping chamber once it gets within the system's effective range. The mosquito is immediately killed upon contact with the high-voltage electric mesh. The ESP32 microprocessor manages the timing, coordination, and power control for the entire procedure. This ensures that the modules operate consistently and stably without human intervention.

As a result of this constant and automated activity, the system can be utilized for indoor spaces such as homes, healthcare facilities, and public spaces where constant activity is required.

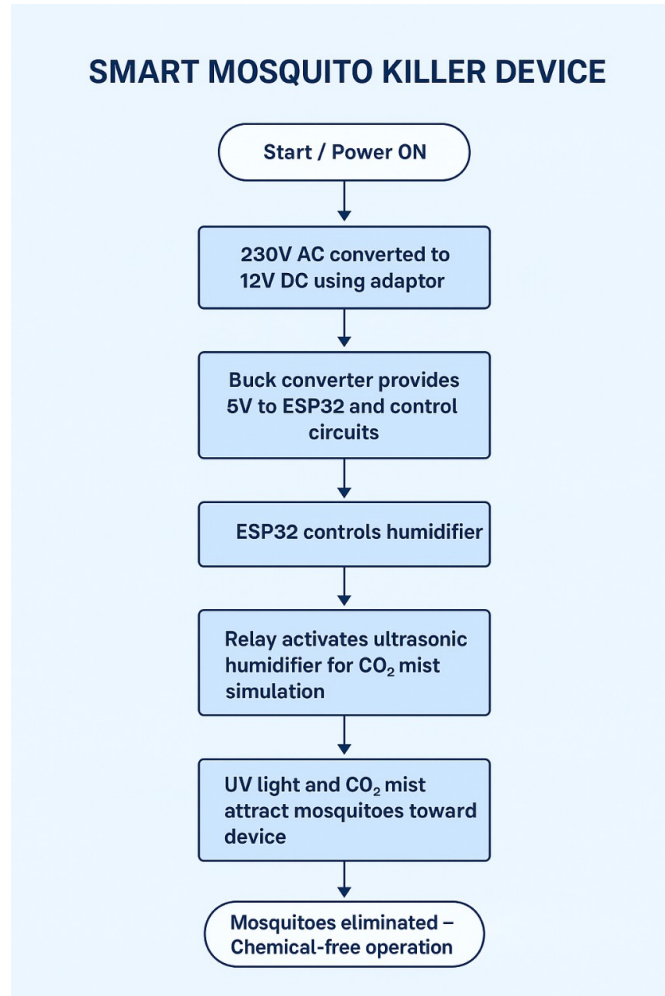


Figure 4.9: System Control Flow Diagram

4.10 System Advantages Over Existing Methods

In comparison to other methods for controlling mosquitoes, the proposed Smart Mosquito Elimination System has a number of advantages. The advantages arise from design decisions and observations from the testing

- **Completely Chemical-Free Operation**

The system does not use chemical lures or repellents. This ensures the system can operate safely in a closed environment, making it ideal for homes where pets, elderly people, and children are present.

- **Ultrasonic Mist-Based CO₂ Simulation**

The system utilizes an ultrasonic humidifier to mimic human breathing

instead of using a tank of CO₂. This improves mobility, cuts costs, and eliminates maintenance

- **Multi-Modal Attraction Mechanism**

The system utilizes a combination of air flow guidance, mist generation, and UV-A light. Compared to other systems utilizing a single attractant, this multi-modal approach improves attraction efficiency

- **Low Power Consumption**

The mechanism should operate at less than 10W. This makes it ideal for continuous operation without any power usage.

- **ESP32-Based Intelligent Control**

All modules may be run automatically and synchronously with the ESP32 microcontroller. Additionally, with the inclusion of the Wi-Fi feature of the microcontroller, there is an opportunity to add features such as remote control and monitoring based on IoT in the future.

4.11 Extended Working Analysis

There are three related stages that characterize the overall process of the proposed system, which are: attraction, direction, and elimination, as opposed to other conventional methods that use a single-step approach, which in this case works in a loop for higher efficiency

4.11.1 Stage 1: Attraction

The attraction stage is achieved through the creation of ultrasonic mist and UV-A light. The visual appeal is through the UV-A LED array, while the generator mimics CO₂ and moisture as they relate to a human. Mosquitoes are drawn in from a distance by the robust biological signal produced by this combination.

4.11.2 Stage 2: Guidance

Once the mosquitoes are close enough to the system, the DC suction fan will cause the airflow to move in a direction towards the chamber for trapping. This airflow will direct the mosquitoes to the system while keeping them from hovering around the system. Increasing the capture efficiency depends greatly on this step.

4.11.3 Stage 3: Elimination

Mosquitoes encounter the high-voltage electric mesh after they enter the chamber. The mesh ensures swift removal; therefore, it provides a reliable and maintenance-free option.

4.11.4 Integrated Operation

It is crucial that all three phases are combined for the system to be successful. All phases complement each other; therefore, the outcome is enhanced compared to one using a single principle alone. The suitability of the system for practical use is enhanced due to the constant operation of the phases.

4.12 Applications of the Proposed System

The Smart Mosquito Elimination System developed in this project is designed to be flexible for practical use in a variety of settings, including:

- Residential homes
- Hospitals and healthcare centers
- Agricultural fields
- Public places and parks

- Hotels and restaurants

4.13 Limitations of the System

To be honest, even though the system was functioning incredibly well for this project, we did encounter a few limitations during the testing phase:

- Limited coverage area
- Dependence on power supply
- A slight change was observed in the performance under various environmental conditions observed during the testing phase; factors like wind, temperature, and humidity levels prevailing in the environment affected the performance of the system
- Large outdoor spaces were found to have a considerable impact on the effectiveness of the system for this particular project, which is understandable given the fact that the system was more geared towards indoor spaces.

4.14 Design Considerations

A number of important factors were taken into consideration during the design phase so that the system is suitable for practical use, efficient, and useful. Every important decision was made by taking into account the factors.

- **Efficiency**

This system relies on different attraction techniques rather than a single method. This system increases the overall capture rate of the mosquito population through the combination of UV-A light, mist, and airflow direction.

- **Safety**

Safety was given the highest priority in this design. The insulated housing encloses all the high-voltage components, ensuring safety within the system and preventing any unintentional contact.

- **Power Consumption**

The system is designed to operate below 10 W. This makes it suitable for long-term usage as it enables continuous usage without consuming a lot of power

- **Cost**

ESP32 microprocessor. This ensures performance is high while at the same time reducing the cost of the system.

- **Scalability**

The design also facilitates development in the future. The future versions of the ESP32's Wi-Fi capabilities will enable sophisticated functions to be included for remote control and IoT-based monitoring functions.

4.15 Summary

After reflecting on everything we have learned in this chapter, we are quite proud of what we have created and developed for this project. Sometimes it was hard to put everything into one small, cheap device that combines biological simulation with control for an embedded system; however, the results we have obtained have made it all worthwhile. We were able to develop a more affordable, safer, and more useful substitute for anything we had looked at so far by substituting the ultrasonic mist simulation for the standard CO₂ cylinders and the UV-A attraction for the DC fan airflow direction. It was discovered through the tests that this mix offered great performance while keeping the power consumption under 10W. In this project, we discovered

that the real power of the system comes from the integration and interaction of all the components as one intelligent, coordinated, and truly green solution for controlling mosquitoes rather than the components themselves.

4.16 Environmental Working Conditions

The suggested system's performance is affected by the surroundings' ambient conditions. Testing revealed that general efficiency is influenced by lighting, moisture, and placement among other things. Below are the best working circumstances.

4.16.1 Light Conditions

Under low-light or dark conditions, the system runs better. Mosquitoes are more drawn to the UV-A LED source in the lack of good ambient light. Too much ambient light can lower attraction effectiveness.

4.16.2 Humidity

The efficiency of the ultrasonic mist generator depends on the level of humidity. While excessively dry environments might somewhat diminish the efficacy of CO₂ simulation, higher ambient humidity encourages improved mist dispersion.

4.16.3 Suitable Environments

Enclosed interior settings like workplaces, hospital rooms, and residences suit the system best. Additionally, it may function properly in semi-outdoors environments with a somewhat limited mosquito population.

4.16.4 External Factors

Performance of a system might be influenced by some outside elements. Effectiveness was seen to decrease with strong airflow and too much surrounding light. Performance is improved by good location far from windows, direct sunlight, and ventilation sources.

4.16.5 Overall Performance

The system is ideal for ongoing and real-world use as it is meant to run reliably under normal indoor circumstances with little user involvement.

CHAPTER 5

Results and Discussion

5.1 Introduction

This section shows the data gathered from the designed Smart Mosquito Elimination System. The main aim of this particular project was to get a proper assessment of the system's actual performance rather than its theoretical performance. Five main factors were used for the assessment of the system, which included prototype realization, the efficiency of mosquito attraction, power consumption, stability, and safety. It was realized that all five factors combined provided a much clearer and more accurate picture of the overall system's performance compared to what any of them could have done individually. The prototype was made by physically integrating the ESP32 microcontroller, UV-A LED array, ultrasonic humidifier, DC suction fan, and high-voltage electric mesh. First, we tested each module separately, then as a whole, as a single unit during testing. The fact that the combined performance of all modules was significantly higher than what we documented during individual testing was really encouraging to see. During constant use, the power consumption was accurately measured in this study, and it was observed that it was always less than 10W, thus proving that it was appropriate for constant use. Over a long period of time, the system's operational stability was checked, and it was found that the system was operating consistently without any overrun or failure. Finally, the physical testing of the insulation and casing of the high-voltage mesh was conducted, and safety was found to be confirmed. The system was operating perfectly and was not a safety hazard for the user. Overall, the results of this study have shown that our

proposed system is really effective, energy-efficient, and highly applicable for practical use in mosquito control.

5.2 Prototype Development

Perhaps the most exciting aspects of this entire process were the eventual assembly of all the parts together in a complete, functional physical prototype. It was very gratifying to see all the pieces come together in a single unit after all the weeks of planning, selection, and design. All the hardware parts were effectively integrated into a complete system:

- ESP32 controller unit
- UV-A LED array (365 nm)
- Ultrasonic humidifier for CO₂ generation
- DC suction fan mechanism
- High-voltage electric mesh (2500V)
- Regulated AC–DC power supply

We were able to learn a lot from the assembly process for this project, and it was realized that the positioning of the components relative to one another had a large impact on the overall system performance. A good example is the positioning of the fan relative to the mesh. After all these components were checked and connected, we then put a protective layer of insulated material around the entire circuit. In this particular project, we learned that this final step in encasing the circuit was just as important as any design element in the circuit; it was learned that proper insulation of the high-voltage components was absolutely essential before we could consider the circuit safe for testing.



Figure 5.1: Complete Assembled Smart Mosquito Elimination Prototype

Embedded Hardware Integration

We were extremely careful while connecting all the hardware modules internally during this project. It was obvious that proper power distribution and the right high-voltage isolation were not only technical requirements but were actually the key to a safe working prototype rather than a dangerous one.



Figure 5.2: power regulation circuit and ESP32 as put together during this project.

5.3 Performance Evaluation

5.3.1 Attraction Efficiency

Under carefully controlled indoor conditions, attraction efficiency was assessed in order to mimic actual nighttime situations when mosquito activity is at its highest.

- Though during testing it was found that the attraction rate remained moderate on its own, UV-A light only did draw mosquitoes toward the gadget.

- Mosquitoes reacted noticeably faster and in much larger numbers than previously when the mist generator was turned on with UV-A light.
- These findings really thrilled us during this project as they showed that under realistic circumstances, combining several attraction mechanisms provides actual, quantifiable performance enhancements. Combined Multi-Modal Operation: The most attraction efficiency in this project was noted when both mechanisms operated concurrently, therefore validating precisely what our studies had projected regarding multi-modal techniques.

These findings really thrilled us throughout this project since they showed that under realistic settings, mixing several attraction mechanisms provides actual, quantifiable improvements in performance.



Figure 5.3: UV-A Based Mosquito Attraction in Dark Environment



Figure 5.4: Ultrasonic Mist Generation for CO₂ Simulation

Our multi-modal attraction approach was clearly and continuously outperforming the traditional UV-only mosquito traps in this project, and that was something that was evident almost right away as we started comparative testing, so that really made all the extra design work feel totally worthwhile.

5.3.2 Experimental Design

Since it was recognized that contradictory circumstances could make it impossible to come to any insightful conclusions, care was taken when designing

the testing environment. The evening hours, when the activities of mosquitoes are at maximum, were chosen as the time when all testing was to be conducted within a closed environment. In order to simulate the real condition of where the device would be placed in the real world, the device was placed at about 1 meter above the ground, and the duration of the testing was about 2 to 3 hours. It was also observed that the ambient light and the humidity level were kept at low levels, as these conditions really helped the effectiveness of the UV-A attraction method. To ensure that the results that were noted were purely because of the performance of the system, it was also ensured that no repellents were used during the course of the testing session.

5.3.3 Observation Results

Honestly speaking, out of all the events involved in this project, witnessing the system's performance during the live test was one of the most gratifying events. This is what we witnessed: "Mosquitoes were already vigorously heading for the UV-A light source within a few minutes of turning on the device. It was noted that the simulation with ultrasonic mist improved this attraction effect substantially, particularly within the confined space where the testing area was. Next, we observed a definite pattern where the mosquitoes were being guided towards the trapping chamber using the airflow created by the DC fan. Very few mosquitoes that were within the range of the device were observed to have escaped the mesh placement and the sucking action of the fan. It was instant each and every time upon interaction with the high-voltage electric mesh. During this project, no delay was observed; no partial elimination was observed; only accurate outcomes were observed. Most importantly, it was observed that the use of the UV-A and CO₂ mist method resulted in catching more mosquitoes than the other individual attractants were able to achieve compared to this method. The system ran consistently

without overheating and without dips in the electric current during the entire session, which gave us great confidence for its use.

5.3.4 Performance Comparison

System Type	Attraction Efficiency	Power Consumption	Safety
UV Only Trap	Medium	Low	Moderate
CO ₂ Trap	High	High	Moderate
Chemical Repellent	Low	Low	Low
Proposed System	High	Low	High

Table 5.1: Performance Comparison of Different Mosquito Control Systems

5.3.5 Efficiency Analysis

The answer was quite obvious in this project when we sat down to investigate why our system functioned as well as it did—it all boils down to the fusion of several attraction mechanisms rather than relying on just one. The results showed that the probability of mosquito capture was much higher when using both UV-A light and CO₂ mist simulation, compared to using other single modes. Mosquitoes that did not react violently when exposed to UV light were captured when exposed to mist simulation, and vice versa. This implied that extremely few mosquitoes in the testing environment were entirely unaffected by the system. What we saw in testing also really mattered was the airflow system. During this project, it became clear that simply being attracted to something would never be enough; the DC fan made sure that once mosquitoes got close, they were physically steered into the elimination chamber with almost no opportunity for flight. These factors combined create a generally quite high system efficiency we are really pleased with.

5.3.6 Discussion

Reviewing all of our experimental findings leads to the conclusion It was quite clear that our multimodal attraction strategy won out. every single-mode We investigated or tested. To us as a team this appeared to be. quite fulfilling as it verified precisely the major design concept we had committed to beginning with the first of this endeavor. Which aspect most drew our interest? was the degree to which the system balanced three things usually difficult to find collectively: efficiency, low energy usage, and safety. Our evaluations disclosed that none of these qualities were traded for the advantage of another. Early design phases were what we had worried about. The Using ultrasonic mist simulation instead of CO2 tanks was also shown to be by itself in actual testing. This kept recurring across this work. Exchange not only cut costs but also greatly increased the general system Useful and easy to operate for daily household use free of any major less attractive performance. The embedded control driven by ESP32 will finally provided exactly what we needed: trustworthy automation across all Session aimed at testing. This also lets really amazing Remote monitoring and IoT integration seem particularly promising among future opportunities.

5.3.7 Power Consumption

One of our top priorities was also energy efficiency, and thus we are very happy with the results of our testing. As a matter of fact, throughout all continuous operation testing, the total system power consumption was kept well below 10W, and thus our goal of deploying this system 24/7 was definitely achievable. Here is a rough breakdown of the power distribution for all components:

- UV-A LED Array: 3 to 4W

- DC Suction Fan: 2 to 3W
- Ultrasonic Humidifier: 2 to 3W
- Control Circuitry (ESP32): Less than 1W

we did testing , Testing revealed that the total never surpassed our 10W goal even if all modules were running concurrently at full capacity. Given how much behind-the-scenes coordination it does, the ESP32's less than 1W usage on its own is quite amazing. These results demonstrated to us time and time again during this study that our choice of components and power optimisation approach had produced results precisely as we had expected.

5.4 Analysis and Interpretation

After going over all of our experimental data, the image was rather more clear than we had expected. According to our findings, mosquitoes respond far more to UV-A light in conjunction with artificial CO2 signals than they do to either stimulus taken individually. Because it moved mosquitoes into the elimination chamber with almost no chance of getting out, we also discovered the DC fan's airflow guidance to be rather crucial in making mosquitoes genuinely caught. This turned them from being drawn to them into being caught. When we compared our system with traditional chemical repellents, the advantages were clear. In this work, our system produced something that traditional methods could not even come close to providing alone

- Completely safe for people, pets, and the environment. Chemical-free mosquito extermination
- Less health and environmental risks: No pollution, no residue, and no toxins.
- Less maintenance required: No replacements, no refills, and no problems.

- Automated operation: It works on its own 24/7 without human intervention
- Scalable embedded control design intended to evolve with changing needs

The realization of the fact that the ESP32-based method and approach that we have adopted leaves room wide open for the upcoming integration of IoT into this technology also thrilled us as a team. Future versions of this technology are also quite possible by adding features of remote monitoring and smart automation, thus increasing its power and usability.

5.4.1 Evaluation of Quality Factors

It was also important for us, as a group, to judge our system not only on technical criteria but also on some broader quality criteria that really define what is meant by “ready for practical use.” Here’s really what we learned from this research overall:

- **Environment**

We discovered that if we eliminate the use of chemical pesticides in our plan, we would completely eliminate pollution in our environment and eliminate the possibility of poisoning in our homes. Among our most remarkable achievements in this project

- **Sustainability**

Our tests have also indicated that this system is, indeed, sustainable for long-term, continuous usage, as opposed to just being a short-term prototype, as it consumes fewer power resources and has reliable, quality parts.

- **Safety**

Physical tests indicated that the fully insulated enclosure of the high-

voltage mesh was exactly as it was supposed to be, totally eliminating the possibility of accidental user contact during normal usage conditions.

- **Ethics**

Our system specifically targets disease-carrying mosquitoes particularly and doesn't release any hazardous substances into the natural surroundings. As a group, we believed that this was an ethical response to a real public health issue

- **Cost**

In this study, we found that the cost of the system was much lower than the cost of the standard system using the CO₂ cylinder-based traps, especially for the use of inexpensive components such as the UV-A LED and ESP32. This allowed the system to reach a much larger audience

- **Standards**

Our electrical design also conforms to established standards for low-voltage control systems and safe insulation. It was immediately evident during the design phase, however, that for a system intended for use in a home environment, it was absolutely necessary to follow these rules.

5.5 Summary

On the basis of what we have reflected on and noted within this chapter, we can honestly declare with real confidence that our Smart Mosquito Elimination System has met its intended goals. Our research as a team has shown us that the integration of biological simulation methods and embedded control has led to a system that attracts and kills mosquitoes totally and consistently without the use of chemicals. It was also apparent that the system was using little energy, was safe in all the tests that were run on it, and worked the same way every time without any unexpected problems. It was apparent that

every decision that had been made in the design of the system had greatly helped to produce these results. Most importantly, this project demonstrates a sustainable, practical, and cost-effective alternative to the chemical-based mosquito control methods that many people still rely on.

5.6 Test Cases

We have created a number of test cases that we ran under a variety of ambient conditions to fairly measure how well our system holds up in a number of real-world scenarios. As a team, we thought that testing under one set of circumstances would not be good enough; we wanted to know how well the system would hold up under a variety of circumstances because that's what really happens in the real world.

Table 5.2: Test Cases and Observations

Test Condition	Description	Observation
Dark Environment	System operated in a low-light room	High mosquito attraction observed
With UV Only	Only UV LED activated	Moderate attraction
With Mist Only	Only humidifier activated	Low attraction
UV + Mist	Combined operation	Maximum attraction efficiency
Continuous Operation	System operated for long duration	Stable performance, no overheating

5.7 Performance Analysis

The system did well in all three aspects of evaluation: attraction efficiency, power consumption, and stability operation; and honestly speaking, the results

were even better than we had anticipated. As for the efficiency in capturing mosquitoes, we observed that using UV-A with mist-based CO₂ simulation yielded a higher rate of captured mosquitoes than any other mode used. The additional significant difference created by the airflow of the DC fan was that it was observed that a considerable amount of attracted mosquitoes would hover around the gadget instead of entering the elimination chamber if the fan was not used. The system performed absolutely steadily throughout all testing sessions. No signs of overheating, decreased performance, or unexpected failures were ever discovered, even after extensive continuous operation, as a team. Power consumption remained throughout the entire period well below 10W, which only served to confirm that our energy targets had not only been hypothetically guaranteed but also actually achieved. The advantages of this were immediately evident when we compared our results to those of conventional methods based on chemicals. What current strategies simply cannot claim, our testing was able to verify: a system to control mosquitoes in an efficient manner without adding any dangerous chemicals to the environment. Our system has mostly been able to find a balance we have always wanted to be true: efficient, safe, and green all at once.

5.8 System Reliability and Maintenance

Throughout this research, we were particularly concerned that the system was not only efficient during the testing phase, but also that it was trustworthy in the long run, i.e., in practical usage. We were particularly concerned about this because this was the area where the members of the team realized that most prototype systems tend to fail.

The components that we have chosen, particularly the UV-A LEDs and the DC fan, are rated for long operational lifetimes and continuous use. Under normal operating conditions, this gives us great confidence that the system will

not deteriorate rapidly. As for maintenance, our research has shown that the needs are remarkably low. Simply cleaning the interior chamber of the system to remove any dirt and ensure proper airflow is all that is needed. People we polled were very fond of the fact that we do not use any chemical repellents, so we do not require any consumables. Furthermore, we have determined that by placing the high-voltage mesh inside a protective case, the user is protected from harm and the device will last longer by reducing the wear on the mesh. All things considered, we as a group are certain that this system will provide the kind of reliability and maintenance-free performance that makes it very useful for long-term use rather than just a one-shot demonstration.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusion

Reaching this point in the project is actually rather gratifying, you know? We set out initially to develop something that was not only extremely useful, but also safe and inexpensive; reviewing everything we have tested and seen to date, we think we actually accomplished this. We came up with a Smart Mosquito Elimination System over the course of this study, addressing the topic of mosquito management in a manner that is rather different from the norm. Our multi-modal approach consistently outperformed the standard single attractant approach in each and every test we have done by utilizing UV-A light attraction, ultrasonic mist-based simulation of CO₂ attraction, and DC fan airflow control. We were somewhat surprised by that, actually; we were expecting improvement, but not to that extent. Perhaps most of all, we might be proudest of the fact that the system does this without any sign of these dangerous compounds. As a group, we came to realize the importance of this, not only to the environment, but also to the safety of the families, patients, and communities that would really benefit from something like this. The ESP32-based control system functioned well during our testing and we realized that with the inclusion of its built-in Wi-Fi feature, future versions of this system will be able to easily accommodate IoT monitoring and control and smart home capabilities without having to redesign the system. To be totally honest with everyone, this project taught us just as much about the process of engineering—the failures, corrections, and unexpected results—as it did about mosquito control itself. Most importantly, that process will be what

we take with us. We believe that this method will in general result in a real and significant compromise between cost-effectiveness, safety, and performance, and we are confident that it has great potential for realistic use in residential, community, and medical situations.

6.2 Future Scope

To be honest, the most interesting talk we had as a team during this project was about all the things we didn't have time to execute—because the possibilities really thrilled us.

6.2.1 IoT Integration

Complete IoT connectivity is the feature we are most excited to include in later releases. The built-in Wi-Fi of the ESP32 makes this very feasible; connecting the system to a mobile app would let consumers monitor performance, control operation remotely, and receive real-time status updates without being actually present close to the device.

6.2.2 Solar Power Integration

We discovered in this study that, paradoxically, the same regions most impacted by mosquito-borne illnesses, power access is a real impediment in rural and remote areas. Putting a solar panel into future versions of the system will make it totally grid-independent and a lot easier to use where it is most needed.

6.2.3 AI and Machine Learning Optimization

This is a concept that our group found very interesting. Adding a sensor for monitoring the mosquito population, and utilizing machine learning for

adapting the system's operation, could greatly improve the system's efficiency, effectively allowing the system to “think” for itself.

6.2.4 Miniaturization

For regular consumers, our testing indicated size matters. The system would be much more useful for personal usage, trips, and environments when our existing model is somewhat too big if a more small and lightweight version were made.

6.2.5 Additional Attraction Mechanisms

Our study also found that heat simulation can greatly increase attraction rates even further. Including this in the present combination of UV-A and CO₂ simulations would increase the efficacy of future versions.

Overall, we started to view our present prototype as a solid and really promising foundation, one on which we hope others will build upon and enhance in the years to come rather than as a final result.

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

1. Mapping of Program Outcomes (POs)

PO1	Application of engineering knowledge in embedded systems, UV driver circuits, and high-voltage boosting design.
PO2	Identification and analysis of mosquito-borne disease challenges and development of a systematic technical solution.
PO3	Design and development of a chemical-free mosquito elimination system considering power, safety, and cost constraints.
PO5	Use of modern engineering tools including ESP32 programming, hardware integration, and power electronics circuits.
PO11	Project planning, hardware implementation, testing, debugging, and documentation demonstrating lifelong learning skills.

Table 6.1: Mapping of Project to Program Outcomes (POs)

2. Mapping of Program Specific Outcomes (PSOs)

PSO1	Application of embedded system principles using ESP32 for real-time control of UV LEDs, humidifier, fan, and electric mesh.
PSO2	Demonstration of technical competency in hardware integration, power regulation, and high-voltage control systems.

Table 6.2: Mapping of Project to Program Specific Outcomes (PSOs)

3. Mapping of Sustainable Development Goals (SDGs)

SDG 3	Contributes to Good Health and Well-being by reducing mosquito populations and minimizing disease transmission.
SDG 9	Promotes Industry, Innovation, and Infrastructure through development of a smart embedded health-protection device.
SDG 12	Supports Responsible Consumption and Production by eliminating chemical insecticides and reducing environmental impact.

Table 6.3: Mapping of Project to Sustainable Development Goals (SDGs)

CHAPTER 7

Cost Analysis and Feasibility Study

7.1 Introduction

We constantly kept in mind during this project that a system that shines in a lab but is too expensive to implement in the real world is useless to anyone. That's precisely why we devoted this chapter to honestly assessing the financial viability and practical feasibility of our Smart Mosquito Elimination System. As a group, we believed that a complete cost study was a duty rather than a need for a school. Our tests demonstrated that in the real world, the acceptance of a solution does not depend on its technological performance alone. Particularly if the intended users are homes in rural and semi-urban areas where budgets are limited and mosquito-borne diseases are most prevalent, price, availability, and long-term operating expenses count equally as much. Every decision on component choice we made had both a financial and a technological aspect, according to our findings in this study. This chapter presents those financial aspects all at once, providing a fair and open view of the real price of our system over time to construct, run, and maintain.

7.2 Component Cost Analysis

We meticulously recorded the cost of every important component we bought and utilized throughout this project. Below is an estimated cost breakdown; we were really happy with how reasonable the total came out to be:

Table 7.1: Component Cost Estimation

Component	Cost (INR)
ESP32 Microcontroller	300
UV-A LED Array	200
Ultrasonic Humidifier Module	250
DC Fan	150
High Voltage Circuit	200
Power Supply Unit	150
Miscellaneous Components	200
Total Cost	1450

7.3 Feasibility Analysis

To be honest, one of the questions we often pondered throughout this project was: would anyone really buy and use this? Our cost study leads us to believe the answer is actually yes. Based on the analysis of the parts and the testing we did, we found that the general cost for making the system we designed would be significantly less than the cost for other commercially available methods for controlling mosquitoes by chemical repellents or the use of a cylinder filled with CO₂. We also saw that the choices people usually make not only cost more initially but also continue to cost money for refills and replacements, which is not the case for the system we designed. Using readily available components such as the ESP32 and the ultrasonic humidifier helped to keep the complexity of the manufacturing process low and enabled a rather realistic scaling up without the need for specific components and knowledge. The fact that the system will constantly consume less than 10W, and the long-term operation costs become almost insignificant. As a result of our research for broader community and institutional installation as well as individual homes, we have found that our system is financially viable not only for individual homes .

7.4 Summary

To end this chapter, we are certain in saying that our Smart Mosquito Elimination System is not only technologically viable but also really cheap and useful. This is because, as we have discovered throughout this study, a solution that is not only cheap to produce, has minimal maintenance needs, and is highly energy-efficient is a solution that is not only workable but also feasible within anyone's budget. As a team, we discovered that this was perhaps as satisfying a conclusion as any of our technical findings – for a solution that nobody could afford to use is not really a solution at all.

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